pandemic would be handled was established in 2020. Rich countries poured resources into the development and testing of vaccines if they had the ability to make them, and if they did not, they began reserving doses of as yet unproven vaccines. Their bets paid off. It turned out that new and old platforms worked well, and vaccines were easy to develop (if not to make) and depending on the platform, worked brilliantly or well. The world's vaccination programme, which had never made more than 4 billion doses of vaccine in any year, made 10 billion doses of SARS-CoV2 vaccines in 2021, and gave at least 9 billion doses. More than half the world's population received at least one dose of a vaccine.

IN 2022, WE MUST REDOUBLE EFFORTS TO STRENGTHEN MANUFACTURE AND DELIVERY SYSTEMS OF VACCINES... IF WE DO NOT REDUCE VIRAL CIRCULATION GLOBALLY, WE DO NOT DECREASE THE RISK OF NEW VARIANTS EMERGING

But there was a terrible chasm between the haves and the have-nots, whether it was with diagnostic tests, sequencing, or vaccines and vaccination. Rich countries began booster doses and vaccinating children before the rest of the world, particularly Africa, received doses to protect their most vulnerable, estimated by the WHO at 20 per cent of the population. In low-income countries, less than 10 per cent of people have received their first dose of vaccine. The COVAX facility, despite advance purchases, was pushed to the back of the queue by more powerful political drivers and failed to meet its 2021 goal of delivering 2 billion doses of vaccine. But there is at least hope now, as the scaled up manufacturing and declining demand from high income countries makes more doses accessible. In 2022, we will have to redouble efforts to strengthen manufacturing and delivery systems to ensure that as much of the world's population is protected as quickly as possible, because of the threat that if we do not reduce viral circulation globally, we do not decrease the risk of new variants emerging.

In a repeat of the experience with alpha in early 2021, we face a new threat in the highly mutated Omicron variant. Omicron's sequence data tell us that we do not yet understand why and how, and perhaps more importantly where and in whom, viruses evolve. This is a highly transmissible virus and it is likely it will outcompete delta to become the dominant strain globally. Omicron may be less severe than previous variants, but it also includes multiple mutations that make it less susceptible to the immune response elicited by prior infection or our current vaccines, making it urgent and essential that we understand the consequences in populations that have received different kinds of vaccines and have different infection histories. Governments have introduced travel bans, curfews and booster doses in attempts to handle this new and unpredictable threat. But it is important to understand that we are in a different place this year than we were at the end of 2020. We know more and have more, in terms of understanding viral behavior and how to use diagnostic tests, drugs and vaccines. Unfortunately, the science of infectious disease epidemiology and public health has been sidelined, even in the medical profession in some places, by inaccurate opinion-driven pronouncement and by political drivers of action.

Without downplaying the work that still remains to be done, or the ways in which we failed and the need to grieve the tremendous losses suffered by many, what has been accomplished in the last two years is nothing short of extraordinary in scale. It can be argued that the mobilisation of science and technology allowed us to understand, track, treat and prevent infections from our viral foe, which while less deadly than other coronaviruses, shows us the shape of things to come in a world where man-made change is creating the potential for an increase in new pandemics. In large measure, we share the blame for the conditions in which pandemics arise, but the huge and historic effort of 2021 shows what our collaborative, collective intellectual resources can achieve.

What do we need? We need to understand that the future Omicron portends is one in which new SARS-CoV2 variants will continue to emerge and some will have evolved sufficiently away from the immune system that we will have new global waves, much like we already experience with influenza. This means that both science and public health need to be at the forefront of our control endeavours, supported by enabling policy. We need the scientific research to tell us which control measures work and which do not, how should testing strategy be used to best protect the vulnerable, what the enablers of community behavior change are, the best clinical protocols for each stage of illness, how to make and test new broadly protective vaccines that do not require a booster every few months, and we need the enabling policy that strategically funds the science at scale and for the long term. These threats are not going away. ■

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